MA3071 - DLI
Financial Mathematics – Section 7
Asset pricing models under equilibrium (Capital asset pricing model)

Dr. Dawei Hao

University of Leicester

Aims of the CAPM

- It considers all investors in the market and extends the mean-variance portfolio theory from Section 6.
- Aim to develop a theory of market prices, driven by the actions of individual investors optimising their portfolios using the Markowitz approach.
- This will be an equilibrium theory: i.e. will derive prices under conditions when supply and demand are balanced. This shows the level that prices (and therefore returns) should shift towards over time.
- The capital asset pricing model (CAPM) was developed in the 1960s by William Sharpe, Jon Lintner and Jan Mossin who won the Nobel prize in Economics for this work.

Assumptions

- All assets are marketable (i.e. can be freely bought and sold, meaning that there will always be buyers for assets that are up for sale and there are no other restrictions).
- Capital markets are perfect:
  - All assets are infinitely divisible, that is, any proportion of assets can be traded;
  - All investors are price takers, i.e. no one investor can influence the market price by buying or selling actions;
  - Taxes and transaction costs do not exist or alternatively do not affect the investment decision;
  - Unlimited borrowing and short selling are allowed, without margin/collateral requirements (i.e. there is no credit risk associated with any borrower);
  - Information is available to every investor at no cost, and all investors possess the same information.

Assumptions, cont.

- A risk-free interest rate (ρ) exists at which all investors can undertake unlimited borrowing or lending. This can be thought of as the rate of interest on bank accounts or the yield on Treasury Bills (both of which would be the same in this idealised market).
- All investors are risk-averse and seek to maximize expected utility over single period time horizons.

Assumptions, cont.

- Investors have homogeneous expectations (and therefore all make the same assumptions):
  - They possess the same investment time horizons and their estimates of the expected returns, variances (or standard deviations), and covariances of returns on risky assets are identical;
  - They all base their portfolio selection decisions on mean-variance optimisation. In other words, all investors perceive investment alternatives and arrive at portfolio decisions in exactly the same manner.

The capital allocation line

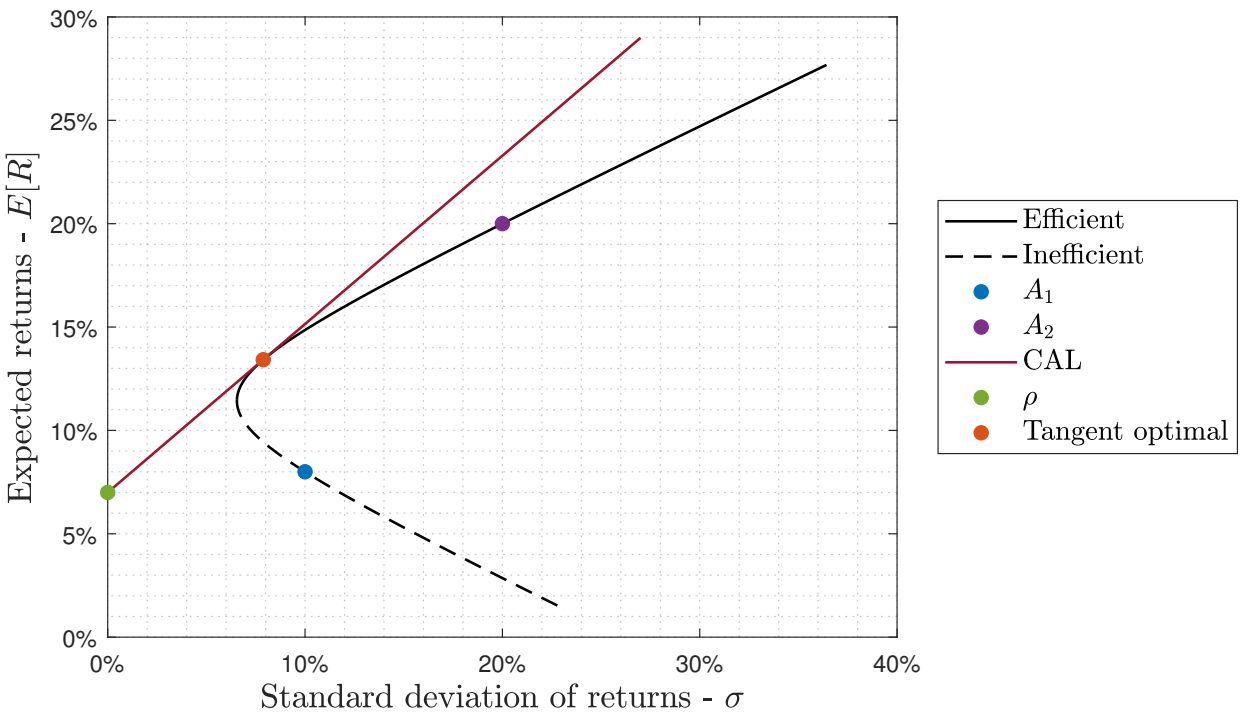
- We want to add the risk free asset to the portfolios that form the efficient frontiers of Markowitz.
- The risk free asset always has zero variance of returns, since it is riskless. It also has zero correlation, and therefore zero covariance with any of the risky assets.
- Notice that adding the possibility of investing in a risk free asset extends the choice of portfolio to a mix of risky assets and the risk free asset.
- Recall the two risky assets from Section 6:

$$A_1 : \sigma_1 = 10\%, \quad \mathbb{E}[R_1] = 8\%$$
$$A_2 : \sigma_2 = 20\%, \quad \mathbb{E}[R_2] = 20\%$$

Assume that the correlation is $c_{12} = -0.5$, and the risk-free interest rate is $\rho = 7\%$.

The capital allocation line, cont.

- Since all investors can borrow and lend at the risk-free rate, the efficient frontier simplifies to a straight line passing through the risk free rate and a unique point on the efficient frontier of risky assets, which is called the **capital allocation line** (CAL).



The capital allocation line, cont.

- The orange point at which the straight line is tangent to the frontier comprised of a portfolio containing only risky assets is called the **tangent portfolio**.
- In the picture above, the tangent portfolio is $(x_1, x_2) = \left(\frac{17}{31}, \frac{14}{31}\right)$ with $\sigma_t = \frac{\sqrt{597}}{310} \approx 0.0788$ and $\mathbb{E}[R_t] = \frac{104}{775} \approx 0.1342$.
- All other portfolios (i.e. those made up of the risky assets only) are inefficient compared to the portfolios including the risk-free asset.

The capital allocation line, cont.

- All investors who agree on the expected returns, risk and covariance of risky assets, regardless of their risk preferences, will choose the same portfolio of risky assets (in this simple example it is the **orange dot**).
- This is called a separation result, because the construction of the risky portfolio and the choice of how much risk to take (asset allocation) are separated.
- Firstly, calculate the optimal risky portfolio, which is a mathematical optimisation problem without reference to the investor's preferences with respect to risk.
- Secondly, choose how to split the amount to be invested between the optimal risky portfolio and the risk free asset (i.e. an asset allocation decision). This depends primarily on the investor's preferences. For this reason, the efficient frontier including the risk-free asset is called the **capital allocation line**.

The capital allocation line, cont.

- The slope of the CAL is:

$$\frac{\mathbb{E}[R_t] - \rho}{\sigma_t}$$

  where $\mathbb{E}[R_t]$ and σ_t are the expected return and standard deviation of return for the tangent portfolio.

- The equation of CAL is:

$$\mathbb{E}[R_p] - \rho = \left(\frac{\mathbb{E}[R_t] - \rho}{\sigma_t} \right) \sigma_p$$

- The slope can be thought of as a risk/reward ratio that shows the additional return required to take on more risk.

The capital allocation line, cont.

- The **higher** this ratio, the **better** for the investor, since they are achieving a better level of additional return per unit of risk.
- In other words, it is the additional return, in excess of the risk-free rate, that investors require before they will accept an additional unit of risk.
- In portfolio analysis, this is also known as the **Sharpe measure**, which is widely used for fund performance assessment purposes.

Find the tangent portfolio

- To determine the tangent portfolio of risky assets, it needs to maximise this ratio, subject to the constraint that the weights invested across all risky assets must sum to 1.

$$\max_{x_1,\cdots,x_n} \quad \frac{\mathbb{E}[R_t] - \rho}{\sigma_t}$$

$$\text{Subject to} \quad \sum_{i=1}^{n} x_i = 1$$

Find the tangent portfolio, cont

- The Lagrangian function is:

$$\mathcal{L}(x_1, \cdots, x_n, \mu) = \frac{\mathbb{E}[R_t] - \rho}{\sigma_t} - \mu \left(\sum_{i=1}^{n} x_i - 1 \right)$$
$$= \frac{\sum_{i=1}^{n} x_i \mathbb{E}[R_i] - \rho}{\sqrt{\sum_{i=1}^{n} x_i^2 \sigma_i^2 + 2 \sum_{i=1}^{n} \sum_{j=i+1}^{n} x_i x_j \sigma_{ij}}} - \mu \left(\sum_{i=1}^{n} x_i - 1 \right)$$

- We then need to solve the following equations:

$$\frac{\partial \mathcal{L}}{\partial x_i} = 0, \quad i = 1, \cdots, n$$
$$\frac{\partial \mathcal{L}}{\partial \mu} = 0$$

Obviously, these equations are non-linear, we can use programming to calculate the weights.

The capital market line and the market portfolio

- If all investors have the same expectations (means, variances and covariances), then the set of efficient portfolios of risky assets and the CAL they use to generate portfolios would be the same. This is because all investors would be using the same expected returns, variances and covariances of returns, which is the situation of **market equilibrium**.
- In equilibrium, the tangent portfolio of risky assets would therefore be the same for all investors. Then the tangent portfolio is called the **market portfolio** and consists of all possible risky assets, weighted by market value (or "**market capitalisation**").
- In practice, the market portfolio is often approximated by broad market indices such as the FTSE100 index in the UK or the S&P500 in the US.

The capital market line and the market portfolio, cont.

- Then, the CAL becomes the capital market line (CML), with equation:

$$\mathbb{E}[R_p] - \rho = \left(\frac{\mathbb{E}[R_M] - \rho}{\sigma_M} \right) \sigma_p$$

  where $\mathbb{E}[R_M]$ and σ_M are the expected return and standard deviation of return for the market portfolio.

- The slope of CML is referred to as the '**market price of risk**', being the 'price' that investors as a whole will charge for every unit of risk that they take on.

- **Remark**: The CML is a special case of the CAL. In equilibrium, all investors will hold the same tangent portfolio called the market portfolio. Then everyone's CAL will be called the CML - effectively the market's consensus of the risk frontier.

Capital asset pricing model

- The CML gives an expression that can be used to relate the return on any efficient portfolio to the market portfolio.
- This can be developed to relate the return on any individual asset to the market portfolio:

$$\mathbb{E}[R_i] - \rho = \frac{Cov(R_i, R_M)}{\sigma_M^2} (\mathbb{E}[R_M] - \rho)$$

  where $\frac{Cov(R_i,R_M)}{\sigma_M^2}$ represents the sensitivity of the expected excess return (over and above the risk-free rate) of asset i to the expected excess market return.
- This equation takes into account the asset's sensitivity to non-diversifiable risk (also known as systematic risk or market risk).

Capital asset pricing model, cont.

- It is common to write the coefficient $\frac{Cov(R_i,R_M)}{\sigma_M^2}$ as β_i, since this is the slope of a linear regression of the expected excess asset return on the expected excess market return. The slope coefficients in regression models are often denoted as β.
- Hence,
$$\mathbb{E}[R_i] = \rho + \beta_i \left(\mathbb{E}[R_M] - \rho\right)$$
This is the classic form of the CAPM that is most commonly encountered.
- **Remark**: In this model, the appropriate measure of risk is no longer the variance of returns, but rather the degree to which the asset contributes to the risk of the market portfolio, as measured by β. In other words, β can be considered a measure of the systematic risk of the asset in question.

Example

In a market where the assumptions of the CAPM hold, there is one risk-free asset (Asset 1) with an annual rate of return of 2.787 and two risky assets with the following properties:

		Rate of return (p.a.)	
State	Probability	Asset 2	Asset 3
1	0.1	5	6
2	0.3	3	3
3	0.4	4	5
4	0.2	6	8

The market capitalisation of Asset 2 is 40,000 and the market capitalisation of Asset 3 is 60,000. The market portfolio is defined by the market capitalisations.

i) Determine the market price of risk.

ii) Calculate the beta of each risky asset.

Shortcomings of CAPM

- The main challenge with the model is twofold. Firstly, there is the issue that the market portfolio that it specifies is not observable since it should contain all possible assets including property, fine art, etc.
- The second problem relates to the many assumptions that the model requires. Taken altogether, these are not realistic. The most important assumption that will not be met is that all investors can borrow and lend at the risk-free rate.

Shortcomings of CAPM, cont.

- Academic studies testing the model have therefore tended to find very little evidence in favour of it. However, they use financial indices as a proxy for the market portfolio so are not applying a fully appropriate test.
- There is a model called the "zero beta" model that seeks to overcome the limitation regarding the risk-free rate. It has this name because it relies on constructing an artificial riskless portfolio using portfolios that are uncorrelated with each other (or have zero β with respect to each other). The form of the model is then very similar to that of the classic model, but the zero beta portfolio replaces the risk-free asset.